

Basic Electrical Engineering

MODULE 4 — PART 2: QUESTION BANK

Module 4: AC Machines — Transformer and Three-Phase Induction Motor

For First-Year Students — with complete answers and diagrams

Course: Basic Electrical Engineering · Course Code: EE100

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 – Q32
Section B	7 marks	Compulsory — from every module	Q1 – Q12
Section C	Numericals	Solved problems (all topics)	N1 – N13

- Every section covers **all the topics** of Module 4 — no topic is left out.
- No question is repeated between sections.
- Section C gives every numerical problem solved step by step, with its diagram.

SECTION A — 2-Mark Questions

Q1. What is a transformer?

Answer: A static machine that transforms power from one circuit to another without changing the frequency; it works on the principle of mutual induction.

Q2. Why is a transformer called a static device?

Answer: Because it has no rotating or moving parts. Hence it has no mechanical (windage or friction) losses.

Q3. What is mutual induction?

Answer: When the current in one coil changes, the flux linking a nearby coil changes and an emf is induced in it; this is mutual induction.

Q4. Define primary and secondary windings.

Answer: The primary winding receives electrical power from the source; the secondary winding delivers the desired output voltage by mutual induction.

Q5. What is a step-up and a step-down transformer?

Answer: A step-up transformer increases the voltage ($N_2 > N_1$); a step-down transformer decreases it ($N_2 < N_1$).

Q6. What is the purpose of the transformer core?

Answer: It provides a low-reluctance path so that the maximum flux produced by the primary is linked with the secondary winding.

Q7. Why is the transformer core laminated?

Answer: To reduce the eddy-current loss in the core.

Q8. State the principle of operation of a transformer.

Answer: An alternating voltage applied to the primary sets up an alternating flux in the core, which links both windings and induces emfs in them by mutual induction.

Q9. What is the no-load current of a transformer?

Answer: The small current I_0 that flows in the primary when the secondary is open-circuited; it sets up the working flux.

Q10. Define the transformation ratio.

Answer: The ratio of secondary to primary turns, $K = N_2/N_1$, which also equals the voltage ratio and the inverse current ratio.

Q11. Write the relation for an ideal transformer.

Answer: For an ideal (100% efficient) transformer, input power = output power, so $V_1 I_1 = V_2 I_2$.

Q12. Write the EMF equation of a transformer.

Answer: $E = 4.44 f N \phi_m$ volts, where ϕ_m is the maximum flux in the core.

Q13. Differentiate between core-type and shell-type transformers.

Answer: In core type the windings surround the two limbs of the core; in shell type the windings surround the central limb. Core type suits high voltage, shell type low voltage.

Q14. What is copper loss in a transformer?

Answer: The ohmic loss in the windings, $I_1^2 R_1 + I_2^2 R_2$. It varies with the load.

Q15. What is hysteresis loss and how is it reduced?

Answer: Loss due to reversal of magnetisation of the core; reduced by using nickel-iron alloys whose hysteresis loop has a small area.

Q16. What is eddy-current loss and how is it reduced?

Answer: Loss due to circulating currents induced in the core; reduced by laminating the core or increasing the resistivity of the core material.

Q17. What are iron or core losses?

Answer: The hysteresis loss and the eddy-current loss, which depend on the magnetic properties of the core material. They are practically constant with load.

Q18. Define the efficiency of a transformer.

Answer: The ratio of output power to input power; it can also be written as $(\text{input} - \text{losses})/\text{input}$. Typical full-load efficiency is 95% to 98.5%.

Q19. Why is the efficiency of a transformer usually found by the losses method?

Answer: Because the output and input are nearly equal (the transformer is highly efficient), so output/input is not accurate; hence $(\text{input} - \text{losses})/\text{input}$ is used.

Q20. State two applications of a transformer.

Answer: Stepping up the voltage for long-distance transmission and stepping it down for distribution and use; also for isolating circuits.

Q21. Name the two main parts of a three-phase induction motor.

Answer: A stationary stator and a revolving rotor.

Q22. What is the stator of an induction motor?

Answer: A steel frame supporting a hollow cylindrical laminated core with evenly spaced slots that hold the stator winding.

Q23. What is a squirrel-cage rotor?

Answer: A rotor with conducting bars laid in slots and shorted at both ends by end rings, forming a cage-like circuit.

Q24. What is a wound (slip-ring) rotor?

Answer: A rotor with a three-phase winding like the stator, usually Y-connected, whose three ends are brought out to three slip rings so the rotor circuit is accessible.

Q25. What is a rotating magnetic field?

Answer: A field of constant magnitude produced by balanced three-phase windings fed from a balanced three-phase supply, which rotates at synchronous speed.

Q26. Give the formula for synchronous speed.

Answer: $n_s = 120 f / P$ rpm, where f is the supply frequency and P the number of poles.

Q27. State the principle of operation of a three-phase induction motor.

Answer: The rotating field induces emf and currents in the short-circuited rotor conductors; these currents interact with the field to produce torque, so the rotor rotates.

Q28. Define slip.

Answer: The fractional difference between synchronous speed and rotor speed: $s = (n_s - n)/n_s$.

Q29. Why can an induction motor never run at synchronous speed?

Answer: If the rotor ran at synchronous speed there would be no relative motion, hence no induced emf, no rotor current and no torque; so it always runs slightly slower.

Q30. How is the rotor frequency related to the supply frequency?

Answer: The rotor frequency is $f_r = s f$, where s is the slip and f the supply frequency.

Q31. On what does the torque of an induction motor depend?

Answer: It is proportional to the product of the flux and the component of rotor current in phase with it: $T \propto \phi I_2 \cos \phi_2$.

Q32. State the applications of a three-phase induction motor.

Answer: Lifts, cranes, hoists, large-capacity exhaust fans, lathe machines, crushers, oil-extracting mills and textile mills.

SECTION B — 7-Mark Questions

Q1. What is a transformer? Explain its construction with a neat sketch.

Answer:

A transformer is a static device that transfers electrical power from one circuit to another without change of frequency, using mutual induction. It consists of two windings — the primary (which takes power from the source) and the secondary (which delivers the output) — wound on a laminated core. The windings are insulated from each other and from the core. The core provides a low-reluctance path and is laminated to reduce eddy-current loss.

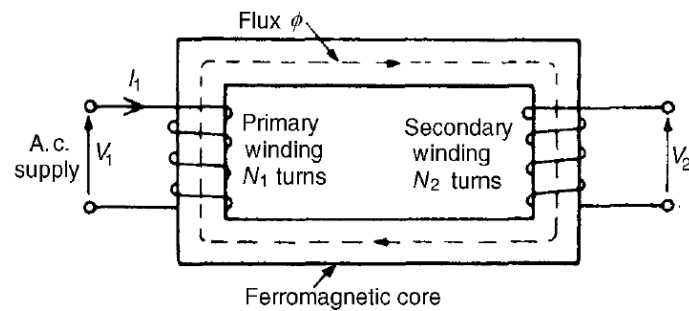


Figure 1: Construction of a transformer.

Single-phase transformer — basic arrangement

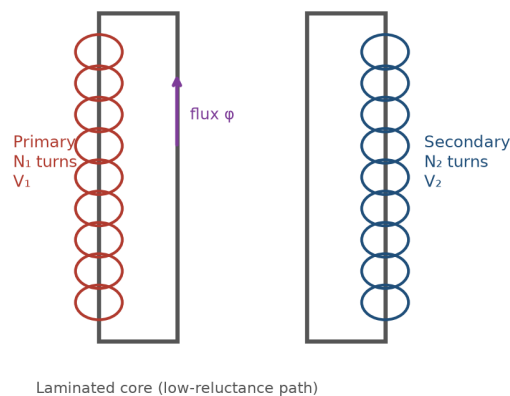


Figure 2: Basic transformer arrangement.

Q2. Explain the working principle of a single-phase transformer.**Answer:**

With the secondary open and an alternating voltage V_1 applied to the primary, a small no-load current I_0 flows and sets up an alternating flux in the core. This flux links both windings and induces emfs E_1 and E_2 . The emf per turn is the same in both windings, so $V_2/V_1 = N_2/N_1 = K$. On load, an ideal transformer satisfies $V_1 I_1 = V_2 I_2$.

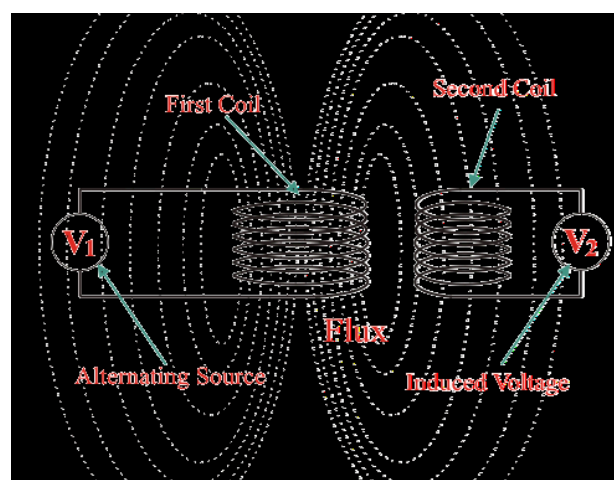


Figure 3: Mutual induction between the windings.

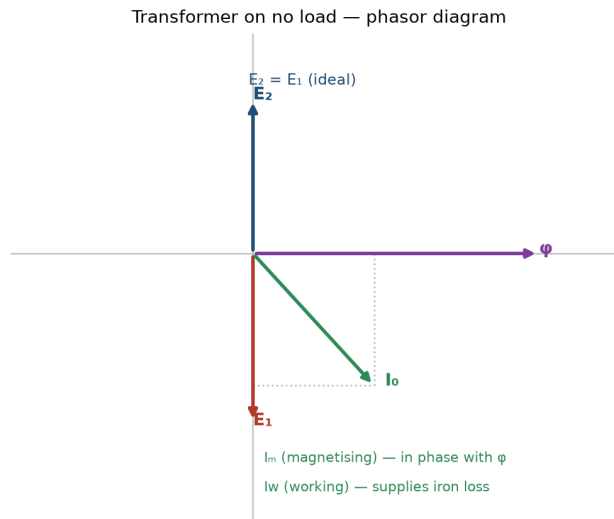


Figure 4: No-load phasor diagram.

Q3. Derive the EMF equation of a transformer and explain the terms.**Answer:**

Let ϕ_m be the maximum core flux and f the supply frequency. The average rate of change of flux is $4f\phi_m$ and the rms value is 1.11 times this, so:

$$E = 4.44 f N \phi_m \text{ volts}$$

where $\phi_m = B_m \times A$, N is the number of turns of the winding and E is its induced emf. Thus $E_1 = 4.44fN_1\phi_m$ and $E_2 = 4.44fN_2\phi_m$.

$$\begin{aligned} \text{Hence rms value} &= \text{form factor} * \text{average value} \\ &= 1.11 * \text{average value} \end{aligned}$$

$$\begin{aligned} \text{Thus rms e.m.f. induced in each turn} &= 1.11 * 4f\phi_m \text{ volts} \\ &= 4.44f\phi_m \text{ volts} \end{aligned}$$

Therefore, rms value of e.m.f. induced in primary,

$$E_1 = 4.44f\phi_m N_1 \text{ volts}$$

and rms value of e.m.f. induced in secondary,

$$E_2 = 4.44f\phi_m N_2 \text{ volts} \quad E_1 \propto N_1, \quad E_2 \propto N_2$$

Figure 5: EMF equation of a transformer.

Q4. Explain the losses in a transformer and how each is reduced.**Answer:**

A transformer has only electrical losses. **Copper loss** is the I^2R loss of the windings ($I_1^2R_1 + I_2^2R_2$) and varies with the load. **Hysteresis loss**, due to reversal of magnetisation, is reduced by using nickel-iron alloys with a small hysteresis loop. **Eddy-current loss**, due to circulating currents in the core, is reduced by laminating the core. The last two together form the iron (core) loss, which is nearly constant with load.

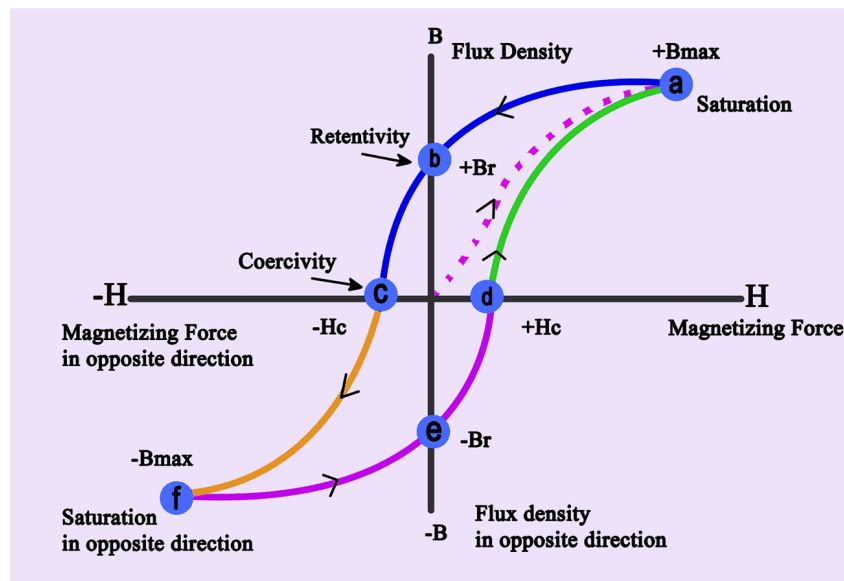


Figure 6: Iron and copper losses.

Q5. Define the efficiency of a transformer and explain how it is calculated.

Answer:

Efficiency is the ratio of output power to input power: $\eta = \text{Output}/\text{Input} = (\text{Input} - \text{Losses})/\text{Input}$. Because a transformer is highly efficient (95% to 98.5% at full load), the input and output are nearly equal, so the losses method is used: $\eta = (\text{Input} - \text{Iron loss} - \text{Copper loss})/\text{Input}$.

Q6. Explain the construction of a three-phase induction motor.

Answer:

It has a stationary stator and a revolving rotor. The stator is a steel frame with a hollow, cylindrical, laminated core having slots for the three-phase stator winding. The rotor is built of punched laminations with slots for the rotor winding, and is of two types: the squirrel-cage rotor (bars shorted at both ends by end rings) and the wound (slip-ring) rotor (a three-phase winding brought out to three slip rings).

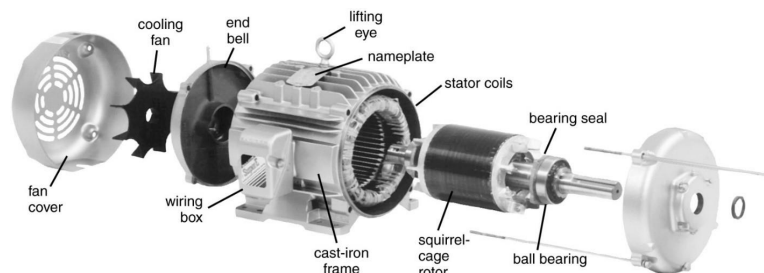


Figure 7: A three-phase induction motor.

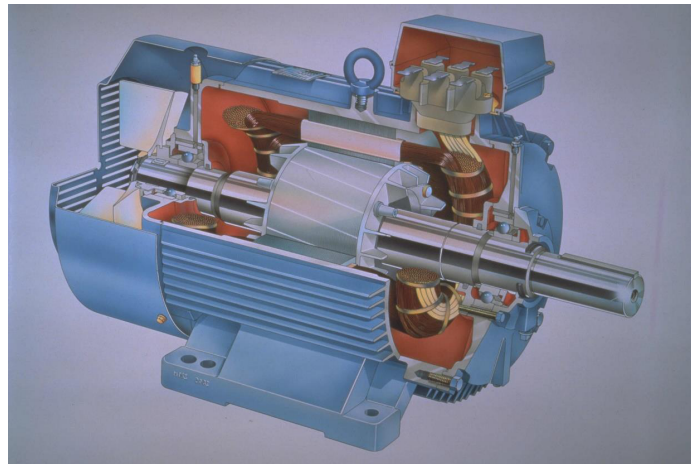


Figure 8: Squirrel-cage rotor.

Q7. Explain the production of the rotating magnetic field and the synchronous speed.

Answer:

Three balanced windings, displaced 120° from each other and fed from a balanced three-phase supply, produce a rotating magnetic field of constant magnitude. The field rotates at the synchronous speed:

$$n_s = 120 f / P \text{ rpm}$$

where f is the supply frequency and P the number of poles.

0°

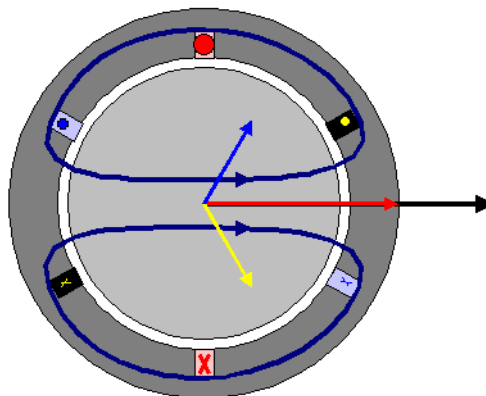


Figure 9: The rotating magnetic field.

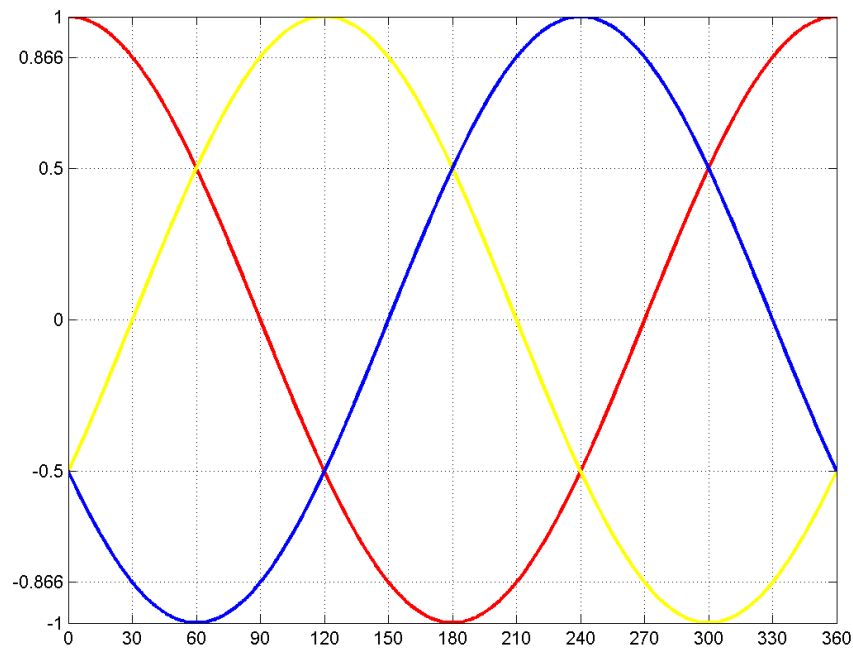


Figure 10: Field interaction with the rotor.

Q8. Explain the principle of operation of a three-phase induction motor.

Answer:

When the supply is switched on, the rotating magnetic field is set up. The relative motion between the field and the stationary rotor conductors induces emf in them (Faraday's law), and since they are short-circuited, currents flow. These currents produce their own fields, which interact with the rotating field to give a net torque, so the rotor accelerates. It settles at a speed slightly below synchronous speed, where the electrical torque equals the load torque.

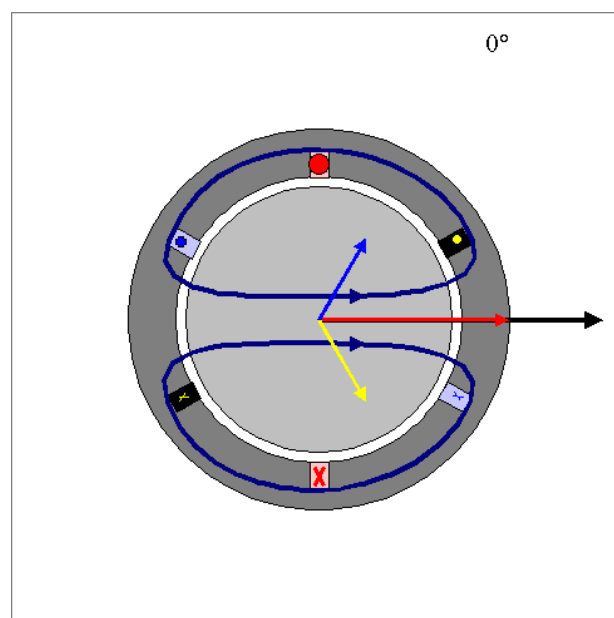


Figure 11: Rotor conductors cutting the rotating field.

Q9. Define slip and explain its importance, with the relation for rotor speed and rotor frequency.

Answer:

Slip is the fractional difference between the synchronous speed and the actual rotor speed:

$$s = (n_s - n)/n_s$$

The rotor speed is $n = (1 - s)n_s$ and the rotor frequency is $f_r = s f$. Slip is about 1% at no load and about 10% at full load. It is essential because without relative motion there would be no induced emf and hence no torque.

Q10. Explain the torque-slip characteristic of an induction motor and its applications.

Answer:

The torque is proportional to the flux and the in-phase component of the rotor current, $T \propto \phi I_2 \cos \phi_2$. It increases with slip up to a maximum (breakdown) value and then decreases. The motor normally runs on the stable, low-slip part of the curve. Applications include lifts, cranes, hoists, exhaust fans, lathe machines, crushers, oil-extracting mills and textile mills.

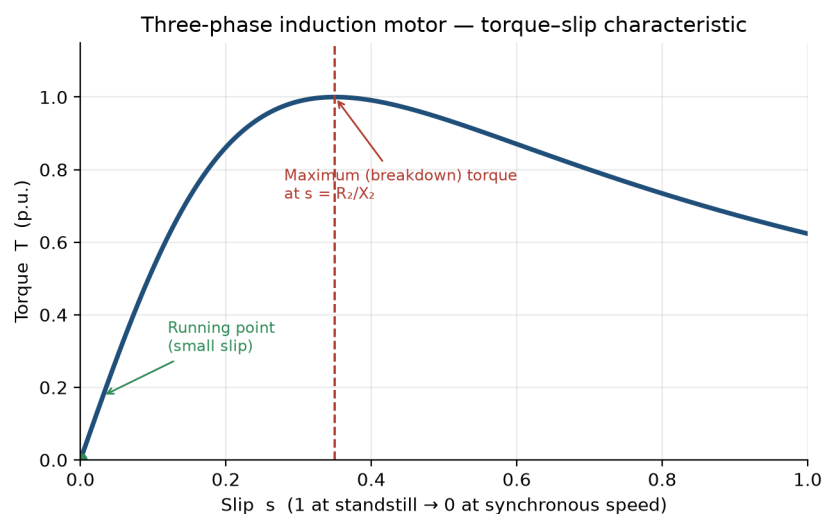


Figure 12: Torque-slip characteristic.

Q11. Compare a squirrel-cage rotor with a wound (slip-ring) rotor.

Answer:

Squirrel-cage rotor	Wound (slip-ring) rotor
Conducting bars shorted by end rings	Three-phase winding like the stator
Cheap, rugged, almost maintenance-free	Costly, needs brushes and slip rings
Low starting torque	High starting torque (external resistance can be added)
Used for most industrial drives	Used where a high starting torque is needed

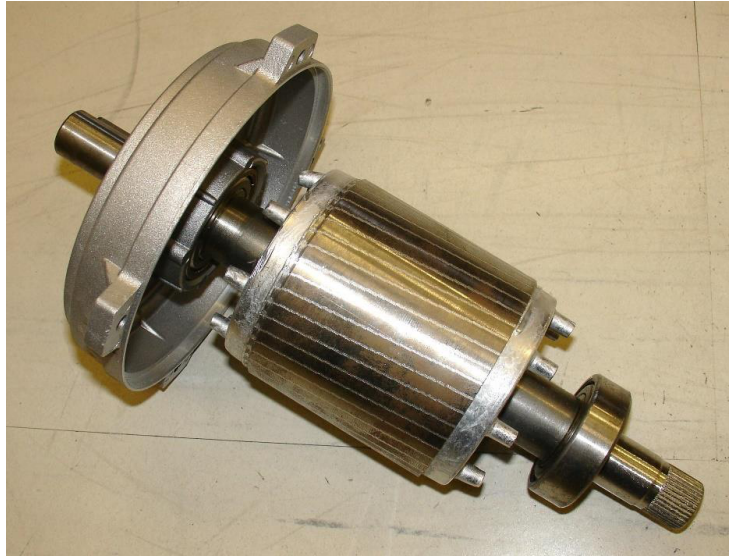


Figure 13: Wound (slip-ring) rotor.

Q12. State the advantages and applications of three-phase induction motors.

Answer:

- Simple and rugged construction, low price and easy maintenance.
- Available in a wide range of ratings (fractional kW to about 10 MW).
- Run at essentially constant speed from no-load to full load.
- Speed depends on the supply frequency, so a variable-frequency drive is used for speed control.
- Used for lifts, cranes, hoists, fans, lathes, crushers, mills and textile machines.

SECTION C — Numerical Problems (Solved)

Every numerical below is solved step by step.

N1. Turns Ratio and Secondary Voltage

Problem: A transformer has 500 primary turns and 100 secondary turns. If 230 V is applied to the primary, find the secondary voltage and state the type of transformer.

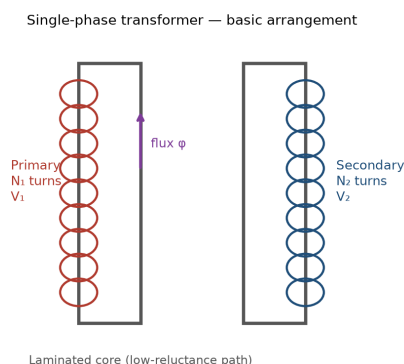


Figure 14: Figure for N1.

Solution:

Transformation ratio $K = N_2/N_1 = 100/500 = 0.2$.

$V_2 = K V_1 = 0.2 \times 230 = 46 \text{ V}$.

Since $V_2 < V_1$, it is a step-down transformer.

Answer: $V_2 = 46$ V (step-down).

N2. EMF Equation of a Transformer

Problem: A transformer has 400 primary turns and a maximum core flux of 0.02 Wb at 50 Hz. Find the primary emf.

$$\begin{aligned}\text{Hence rms value} &= \text{form factor} * \text{average value} \\ &= 1.11 * \text{average value}\end{aligned}$$

$$\begin{aligned}\text{Thus rms e.m.f. induced in each turn} &= 1.11 * 4f\Phi_m \text{ volts} \\ &= 4.44f\Phi_m \text{ volts}\end{aligned}$$

Therefore, rms value of e.m.f. induced in primary,

$$E_1 = 4.44f\Phi_m N_1 \text{ volts}$$

and rms value of e.m.f. induced in secondary,

$$E_2 = 4.44f\Phi_m N_2 \text{ volts}$$

$$E_1 \quad N_1$$

Figure 15: Figure for N2.

Solution:

$$E_1 = 4.44 f N_1 \phi_m.$$

$$E_1 = 4.44 \times 50 \times 400 \times 0.02 = 1776 \text{ V.}$$

Answer: $E_1 = 1776$ V.

N3. Turns from Core Area and Flux Density

Problem: A 10 kVA, 3300/240 V, 50 Hz, single-phase transformer has a core area of 300 cm² and a flux density of 1.3 T. Find (i) the primary turns, (ii) the secondary turns and (iii) the primary full-load current.

Solution:

$$\text{Maximum flux } \phi_m = B_m \times A = 1.3 \times 300 \times 10^{-4} = 0.039 \text{ Wb.}$$

$$(i) N_1 = E_1 / (4.44 f \phi_m) = 3300 / (4.44 \times 50 \times 0.039) = 381 \text{ turns.}$$

$$(ii) N_2 = E_2 / (4.44 f \phi_m) = 240 / (4.44 \times 50 \times 0.039) = 28 \text{ turns.}$$

$$(iii) I_1 = \text{kVA} \times 1000 / V_1 = 10000 / 3300 = 3.03 \text{ A.}$$

Answer: $N_1 = 381$, $N_2 = 28$, $I_1 = 3.03$ A.

N4. Primary and Secondary Full-load Currents

Problem: A 50 kVA, 2400/240 V single-phase transformer supplies full load. Find the primary and secondary full-load currents.

Solution:

$$I_1 = \text{kVA} \times 1000 / V_1 = 50000 / 2400 = 20.83 \text{ A.}$$

$$I_2 = \text{kVA} \times 1000 / V_2 = 50000 / 240 = 208.33 \text{ A.}$$

Answer: $I_1 = 20.83$ A, $I_2 = 208.33$ A.

N5. Copper Loss of a Transformer

Problem: The primary of a transformer carries 5 A through a resistance of 0.4 Ω , and the secondary carries 50 A through a resistance of 0.005 Ω . Find the total copper loss.

Solution:

$$\begin{aligned}\text{Copper loss} &= I_1^2 R_1 + I_2^2 R_2 \\ &= 5^2 \times 0.4 + 50^2 \times 0.005 = 10 + 12.5 = 22.5 \text{ W.}\end{aligned}$$

Answer: Copper loss = 22.5 W.

N6. Efficiency from Output and Losses

Problem: A transformer delivers an output of 10 kW and has total losses of 500 W. Find its efficiency.

Solution:

$$\begin{aligned}\eta &= \text{Output} / (\text{Output} + \text{Losses}). \\ \eta &= 10000 / (10000 + 500) = 10000 / 10500 = 0.9524. \\ \eta &= 95.24\%.\end{aligned}$$

Answer: $\eta = 95.24\%$.

N7. Efficiency from Input and Losses

Problem: A transformer takes an input of 5 kW and the total losses are 250 W. Find its efficiency.

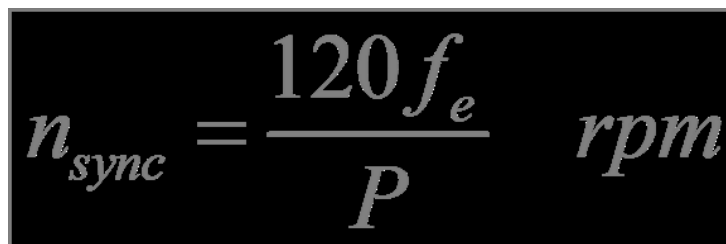
Solution:

$$\begin{aligned}\eta &= (\text{Input} - \text{Losses}) / \text{Input}. \\ \eta &= (5000 - 250) / 5000 = 4750 / 5000 = 0.95 = 95\%.\end{aligned}$$

Answer: $\eta = 95\%$.

N8. Synchronous Speed of an Induction Motor

Problem: A three-phase induction motor is connected to a 50 Hz supply and has 4 poles. Find its synchronous speed.



$$n_{sync} = \frac{120 f_e}{P} \text{ rpm}$$

Figure 16: Figure for N8.

Solution:

$$\begin{aligned}n_s &= 120 f / P. \\ n_s &= 120 \times 50 / 4 = 1500 \text{ rpm.}\end{aligned}$$

Answer: $n_s = 1500 \text{ rpm.}$

N9. Slip of an Induction Motor

Problem: A 4-pole, 50 Hz induction motor runs at 1440 rpm. Find its slip.

Solution:

$$\begin{aligned}n_s &= 120 \times 50 / 4 = 1500 \text{ rpm.} \\ s &= (n_s - n) / n_s = (1500 - 1440) / 1500 = 0.04.\end{aligned}$$

Answer: Slip $s = 0.04$ or 4%.

N10. Rotor Speed from Slip

Problem: A 6-pole induction motor is supplied at 50 Hz and runs with a slip of 5%. Find the rotor speed.

Solution:

$$n_s = 120 \times 50 / 6 = 1000 \text{ rpm.}$$

$$n = (1 - s)n_s = (1 - 0.05) \times 1000 = 950 \text{ rpm.}$$

Answer: $n = 950$ rpm.

N11. Rotor Frequency

Problem: A 50 Hz induction motor runs with a slip of 4%. Find the frequency of the rotor current.

Solution:

$$f_r = s f.$$

$$f_r = 0.04 \times 50 = 2 \text{ Hz.}$$

Answer: $f_r = 2$ Hz.

N12. Number of Poles

Problem: A three-phase induction motor has a synchronous speed of 1000 rpm on a 50 Hz supply. Find the number of poles.

Solution:

$$n_s = 120 f / P, \text{ so } P = 120 f / n_s.$$

$$P = 120 \times 50 / 1000 = 6 \text{ poles.}$$

Answer: $P = 6$ poles.

N13. Torque-Slip Reasoning (Numerical)

Problem: An induction motor has a synchronous speed of 1500 rpm. At a certain load its slip is 4%. Find (i) the rotor speed and (ii) the speed at which the relative speed between the field and rotor is zero.

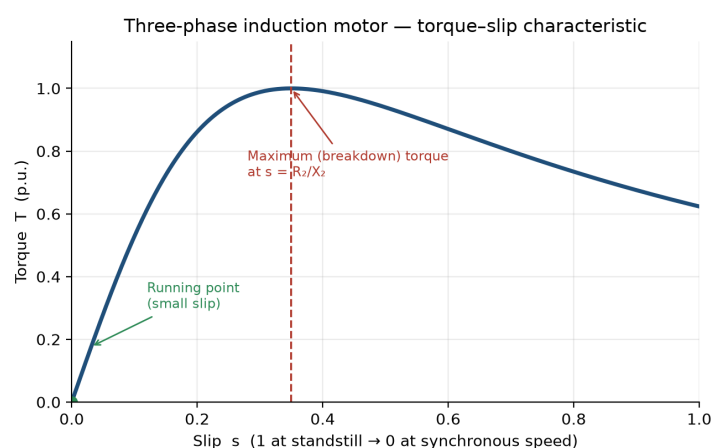


Figure 17: Figure for N13.

Solution:

(i) $n = (1 - s)n_s = (1 - 0.04) \times 1500 = 1440 \text{ rpm.}$

(ii) The relative speed is zero only at synchronous speed, i.e. 1500 rpm.

But at synchronous speed no torque is produced, so the motor cannot run there.

Answer: (i) $n = 1440$ rpm; (ii) synchronous speed = 1500 rpm (no torque).